

Hamza Ehsan

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WORK EXPERIENCE

AMAZON WEB SERVICES | Software Development Engineer

Aug 2022 - Present | Seattle, WA

- Migrated critical APIs from a parent service to a team owned microservice which helped reduce development time for new product features.
- Created an automated integration test suite which supported the CI/CD pipeline for a new service which helped in service stabilization and a seamless product rollout.
- Worked on creating an AWS Fargate based microservice in Java to migrate critical APIs from a legacy monolith service without impacting customer traffic.
- Expanded AWS Connect Flows to a new region through LPT deployment to create a more rigid fail safe environment for customers in case of a large-scale regional outage of service.

NINJATRADER | Full-Stack Engineer

Feb 2022 - Aug 2022 | Chicago, IL

- Led the redesign of the onboarding platform in React using Redux, Recoil, and Material UI.
- Migrated critical reporting metrics from deprecated service to our platform with no downtime, reclaiming 2TB storage and massively decreasing operational load.
- Integrated mission-critical APIs during the acquisition of the Tradovate trading platform which helped migrate 100,000 trading customers to NinjaTrader's new platform.

WEIDERT GROUP | Software Engineer

Aug 2020 - Feb 2022 | Chicago, IL

- Created a custom CSS minified tool that reduces app bundle sizes by 20%. Built using Python and flask on average increasing first-load times by 8%.
- Designed and developed a new backend analytics dashboard in React to handle critical forecasting metadata and monitor API interactions and uptime.
- Worked across the entire stack including API, administrator tools, and relational databases.

SELECTED PROJECTS

DroneSeek | Autonomous Facial Recognition Drone

- Created flight software for an autonomous facial recognition drone that relies on computer vision to navigate indoors using a Tello drone, OpenCV, Pandas, Pygame, and Flask.

Texties | SMS note-taking solution

- Engineered a full-stack SMS-based note-taking app using Twilio, Flask, Next.js, Netlify, and Heroku. Integrated a custom-built multi-factor authentication to reduce overhead.

EDUCATION

Lawrence University

Bachelors of Computer Science
Aug 2016 - Nov 2020 | Appleton, WI

SKILLS

Languages

JavaScript • Python • Java • Ruby

Frameworks

React • Git • PostgreSQL
Flask • NodeJS

Tools

Visual Studio Code • AWS
Google Cloud Services • Docker

AWARDS

MakeHarvard

Harvard University (2019)

DifferenceMaker

U. of Massachusetts (2019)

The Pitch

UW – Green Bay (2019)

LuLaunch

Lawrence University (2018)

MadHacks

UW – Madison (2018)

COURSEWORK

Computer Science

Systems programming • Data Structures • Algorithm Analysis
Theory of Computation • Computer Graphics

Data Science

Machine Learning • Neural Networks • Data Visualization
Data Scientific Programming